Multi-Stage BJT Amplifier Design (2025)

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Abstract*— This experiment aimed to design and analyze a two-stage amplifier utilizing 2N2222A bipolar junction transistors (BJTs) in a common-emitter configuration. The circuit was designed to achieve a specified voltage gain while maintaining signal integrity, utilizing capacitive coupling between stages to ensure proper AC signal transmission and minimize DC offset. The design considerations included establishing a stable Q-point through appropriate DC biasing, selecting resistor values to balance gain and thermal stability, and optimizing impedance matching through an emitter follower stage. The amplifier's performance was evaluated through theoretical calculations, simulations, and experimental measurements. Key parameters such as voltage gain, input impedance, and output impedance were analyzed and compared against expected values, demonstrating the circuit's ability to meet design requirements. The results confirmed that the amplifier achieved the desired gain while maintaining signal fidelity, validating the design methodology and component selection.

I. INTRODUCTION

Amplifiers are essential in electronic circuits, enabling the amplification of weak input signals for further processing. This experiment focuses on designing a two-stage amplifier using 2N2222A bipolar junction transistors (BJTs) in a common-emitter configuration to achieve a specified voltage gain while maintaining signal integrity. The circuit employs capacitive coupling between stages to block DC offset and ensure proper AC signal transmission.

The design constraints include a fixed supply voltage of V_{CC}=15V, ensuring the transistors operate in the active region with appropriate DC biasing. The biasing network must establish a stable Q-point, minimizing variations due to transistor parameter tolerances. The voltage gain must meet design specifications while preventing excessive loading effects between stages. Additionally, the selection of R_E and R_C must balance gain stability and thermal stability, with R_E providing local negative feedback to reduce gain variability. Coupling capacitors C_I, C_B, and C_O must be chosen to allow proper frequency response while avoiding excessive signal attenuation.

This post-lab report outlines the circuit design, including component selection and theoretical analysis, to predict expected performance characteristics before implementation.

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II. SYSTEM DESCRIPTION

The designed circuit is a two-stage amplifier utilizing 2N2222A bipolar junction transistors (BJTs), which can be seen in Figure 1. The first stage is a **common-emitter amplifier**, responsible for voltage gain, while the second stage is an **emitter follower** (also known as a common-collector amplifier), which provides impedance matching and current gain without additional voltage amplification. While the common emitter provides a large enough voltage gain, it is impossible to have an output resistance below 8 Ω with the given constraints. This configuration ensures high overall gain while maintaining a low output impedance to effectively drive the load.

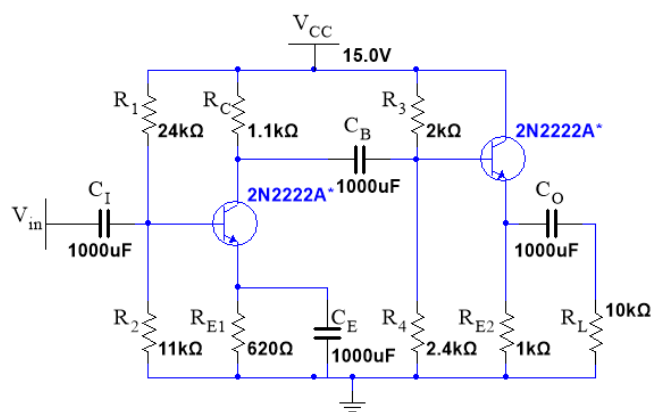

Figure 1: The Multi-Stage BJT Amplifier Circuit Design

A. Common-Emitter Voltage Amplifier

The first stage is a common-emitter amplifier designed to provide voltage gain. The biasing network consists of resistors R1 and R2, forming a voltage divider to establish a stable base voltage, ensuring the transistor operates in the active region. The emitter resistor RE provides local negative feedback, which stabilizes the operating point and reduces distortion. The gain of this stage is approximately:

$$A_v \approx \frac{R_C}{R_E}. \quad (1)$$

Where R_C is the collector resistor, and R_E is the effective emitter resistance (which is modified by the bypass capacitor C_E at high frequencies). Capacitor C_I at the input prevents DC biasing interaction with the signal source, while coupling capacitor C_B transfers the amplified AC signal to the next stage without affecting the DC biasing of the emitter follower.

B. Emitter Follower (Buffer Stage)

The second stage is an emitter follower, which provides high input impedance and low output impedance, making it ideal for driving the load without significantly loading the first stage. The voltage gain of this stage is close to unity:

$$A_v \approx 1. \quad (2)$$

Resistors R3 and R4 form the biasing network, like the first stage, ensuring the transistor remains active. The emitter resistor RE2 helps set the DC operating point and provides some negative feedback, improving linearity. Since the voltage gain is approximately one, the main purpose of this stage is to deliver current gain and impedance matching to drive the load RL. The final output is passed through coupling capacitor CO, which blocks any DC component before reaching the load.

- Impedance Considerations and Frequency Response
- High Input Impedance: The emitter follower's high input impedance prevents excessive loading of the first stage, preserving voltage gain.
- Low Output Impedance: The low output impedance of the emitter follower allows efficient signal transfer to the load.

Frequency Response: The capacitors CI, CB, and CO act as high-pass filters, setting the lower cutoff frequency of the amplifier. The emitter bypass capacitor CE enhances gain by effectively reducing the emitter resistance at high frequencies.

III. Experimental Evaluation

A. Performance Metrics

R_{out} can be found utilizing (1):

$$R_{out} = \frac{1}{g_m} + \left(\frac{R_C \| R_3 \| R_4}{\mathrm{b}+1}\right) \Big\| R_{E2} \leq 8 (1)$$

If (2) is true, then (1) will be true:

$$\frac{1}{g_m} + \left(\frac{R_C \| R_3 \| R_4}{b+1}\right) \leq 8 \quad (2)$$

If (3) and (4) are true, then (2) will be true:

$$\frac{1}{g_m} = \frac{V_T}{I_{C2}} \leq 4 \quad (3)$$

$$\left(\frac{R_C \| R_3 \| R_4}{\mathrm{b}+1}\right) \leq 4 \quad (4)$$

Utilizing (3), (5) I_{C2} can be determined to be:

$$I_{C2} \geq 0.0065 \quad (5)$$

Based on (5), an I_C = 7.5 mA is chosen. Utilizing (4), (6)-(8) can be determined:

$$R_C \| R_3 \| R_4 \leq 4(\mathrm{b}+1) \quad (6)$$

$$R_C \leq 8(\mathrm{b}+1) \quad (7)$$

$$R_3 \| R_4 \leq 8(\mathrm{b}+1) \quad (8)$$

Utilizing β = 150, (9) is calculated:

$$R_C \leq 8(150+1) = 1208 \quad (9)$$

From the fact that higher values increase the voltage gain of the common emitter, a R_C = 1,100 Ω was chosen. For maximum symmetry, (10) is ideal:

$$V_{CE} = V_E \quad (10)$$

With V_{CC} = 15V, V_E should be equal to 7.5V. With the assumed 7 mA, (11) can be calculated:

$$R_{E2} = \frac{V_E}{I_E} = \frac{V_E(\beta+1)}{I_E(\beta)} \quad (11)$$

This leads to (12):

$$R_{E2} = \frac{7.5(150+1)}{0.0075(150)} = 1006.\bar{6} \quad (12)$$

In the lab a nominal R_{E2} = 1000 Ω will be used as it is the closest value below the calculated one. Increasing the resistance will decrease the current, which is unwanted as the current chosen is toward the bottom of the range. To determine R_3 and R_4, (13)-(15) are necessary. The effect of R_E is negligible due to it being multiplied by β, and the resistance being much greater than R_3 and R_4:

$$V_{B2} = V_{E2} + V_{BE2} \quad (13)$$

$$V_{B2} = V_{CC} \frac{R_4}{R_3+R_4} \quad (14)$$

With V_{BE2} assumed to be 0.7V, (13) can be utilized to find (15):

$$V_{B2} = 7.5 + 0.7 = 8.2 \quad (15)$$

With V_{B2} calculated, (14) gives (16):

$$\frac{R_4}{R_3+R_4} = \frac{8.2}{15} = 0.54\bar{6} \quad (16)$$

Utilizing (8), (17) can be calculated:

$$R_3 \| R_4 \leq 8(150+1) = 1208 \quad (17)$$

This in conjunction with (16) gives us (18):

$$0.54\bar{6} R_3 \leq 1208 \quad (18)$$

This means $R_3 \leq$ 2,209.75. Plugging this into (16) gives us (19):

$$\frac{R_4}{2209.75+R_4} \leq 0.54\bar{6} => R_4 \leq 2664.71 \quad (19)$$

The voltage gain for the common emitter in the circuit is seen as (20)

$$|A_V| = \frac{R_3 \| R_4 \| R_C}{\frac{1}{gm} + R_{E1} + \frac{R_1 \| R_2}{\beta+1}} = \frac{I_{C1}(R_3 \| R_4 \| R_C)}{V_T} \geq 100 \quad (20)$$

To conform with the ratio between R_3 and R_4 as well as promote a high voltage gain of the common emitter, the values $R_3 = 2{,}000\ \Omega$ and $R_4 = 2{,}400\ \Omega$. Utilizing (20), a minimum I_{C1} can be found as seen in (21):

$$\frac{I_{C1}(2000 \| 2400 \| 1100)}{0.026} \geq 100 \Rightarrow I_{C1} \geq 0.00473 \quad (21)$$

R_{in} of the circuit is calculated with (22):

$$R_{in} = R_1 \| R_2 \| r_\pi \geq 600 \quad (22)$$

Given the max source current of 20 mA and that R_1 and R_2 are substantially greater than r_π, (24) can be derived:

$$r_\pi = \frac{b(V_T)}{I_{C1}} \geq 600 \quad (23)$$

This results in (24):

$$I_{C1} \leq \frac{150(0.026)}{600} = 0.0065 \quad (24)$$

To ensure the input impedance, an $I_{C1} = 6$ mA was chosen. To ensure the highest degree of symmetry, (25) should be true:

$$V_{CE1} = V_{R_C} = I_{C1} R_C = 0.006(1100) = 5.5 \quad (25)$$

Utilizing (25), V_{E1} is calculated to be 4 V. (26) is then used to find R_{E1}:

$$R_{E1} = \frac{V_{E1}(\beta)}{I_{C1}(\beta+1)} = \frac{4(150)}{0.006(151)} = 662.25 \quad (26)$$

In the lab a nominal $R_{E1} = 620\ \Omega$ will be used as it is the closest value below the calculated one. Increasing the resistance will decrease the current, which is unwanted as the current chosen is toward the bottom of the range. To calculate R_1 and R_2, (27) and (28) can be utilized:

$$V_{B1} = V_{E1} + V_{BE1} \quad (27)$$

$$V_{B1} = V_{CC} \frac{R_2}{R_1 + R_2} \quad (28)$$

With V_{BE1} assumed to be 0.7V, (27) can be utilized to find (29):

$$V_{B1} = 4 + 0.7 = 4.7 \quad (29)$$

(30) should be true. Utilizing (30), (31) can be calculated:

$$I_{R_1} = I_{R_2} \geq 10 I_{B1} = 10\left(\frac{I_{C1}}{\beta}\right) \quad (30)$$

$$I_{R_1} = I_{R_2} \geq 10\left(\frac{0.005}{150}\right) = 0.000\overline{3} \quad (31)$$

From this (32) can be calculated:

$$R_2 = \frac{V_{R_2}}{I_{R_2}} \leq \frac{4.7}{0.000\overline{3}} = 14{,}100 \quad (31)$$

Utilizing (28), (32) can be calculated:

$$\frac{R_2}{R_1 + R_2} = \frac{4.7}{15} = 0.31\overline{3} \quad (32)$$

This can then be utilized in (32) to achieve (33)

$$\frac{14100}{R_1 + 14100} \leq 0.31\overline{3} \Rightarrow R_1 \leq 30{,}900 \quad (32)$$

Wanting a high input resistance, a high value for R_1 and R_2 should be chosen while adhering closely to the ratio between the two values to ensure the expected V_{B1}. The chosen values are $R_1 = 24{,}000\ \Omega$ and $R_2 = 11{,}000\ \Omega$.

B. Verification

To ensure the validity of the calculated values, the restraints were proven to be met by reusing the values in the relevant equations seen in (33)-(37)

$$|A_V| = \frac{I_{C1}(R_3 \| R_4 \| R_C)}{V_T} = \frac{6(2 \| 2.4 \| 1.1)}{0.026} = 126.92 \geq 100 \quad (33)$$

$$R_{in} = R_1 \| R_2 \| \frac{b(V_T)}{I_{C1}} = 24000 \| 11000 \| \frac{150(0.026)}{0.006} \geq 600 \quad (34)$$

$$R_{out} = \frac{V_T}{I_C} + \left(\frac{R_C \| R_3 \| R_4}{b+1}\right) \Big\| R_{E2} = \frac{0.026}{0.0075} + \left(\frac{1100 \| 2000 \| 2400}{150+1}\right) \Big\| 1000 = 7.094 \leq 8 \quad (35)$$

For the emitter follower stage, as stated previously is slightly less than 1 but is negligible due to the high β values that are assumed for this lab (between 150 and 250). Even with a voltage gain of 0.9 for the emitter follower, by multiplying the two voltage gains the total voltage gain is 114.23, well above the threshold.

The circuit was simulated in MultiSim with a β walue of 150 due to this being the smallest beta value in the range, meaning it would have the greatest impact on the results of the simulation.. The input impedance of the circuit was shown to be above 600 within the frequency range. This can be seen in the y2 value in Figure 2.

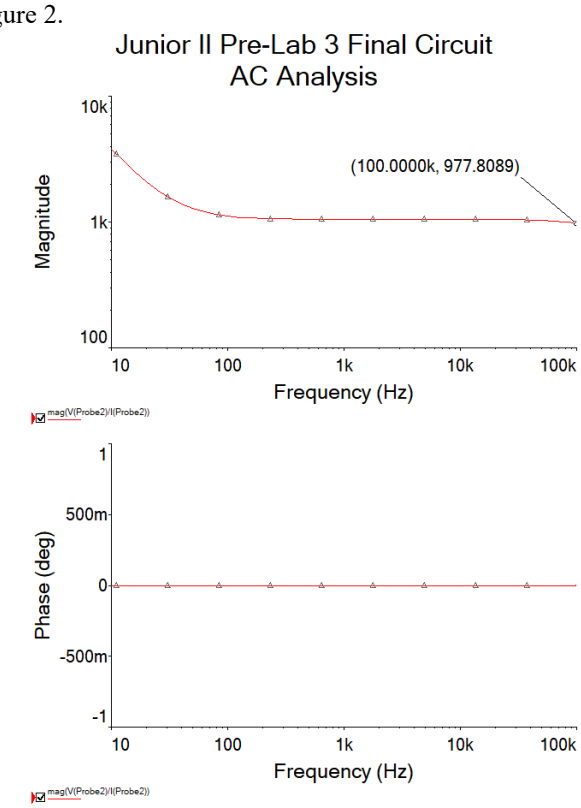

Figure 2: The Input Impedance (R_{in}) Bode Plot

The voltage gain was found to be well above the 100 minimum requirement, with its gain around 116. The mid-band gain was found to be well above 100Hz as can be seen in Figure 3, with a β = 150 and Figure 4, with a β = 250.

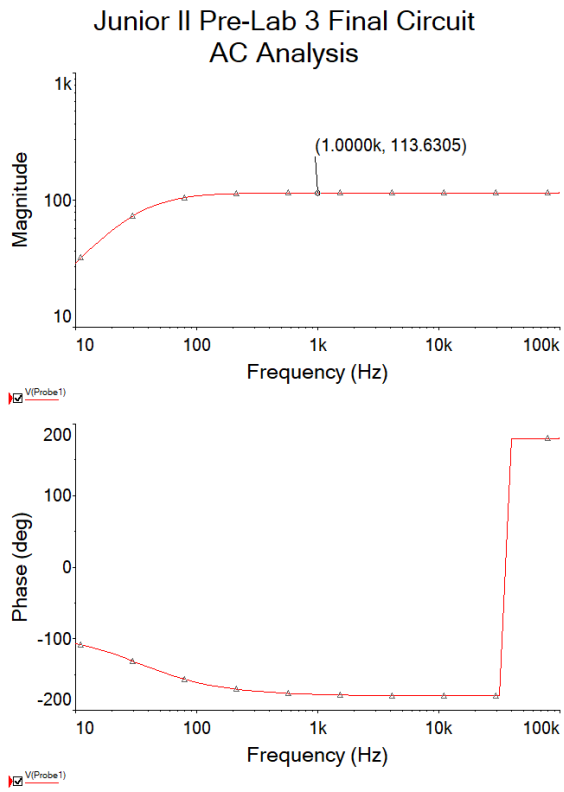

Figure 3: The Voltage Gain $|A_{VO}|$ Bode Plot with a $\beta = 150$

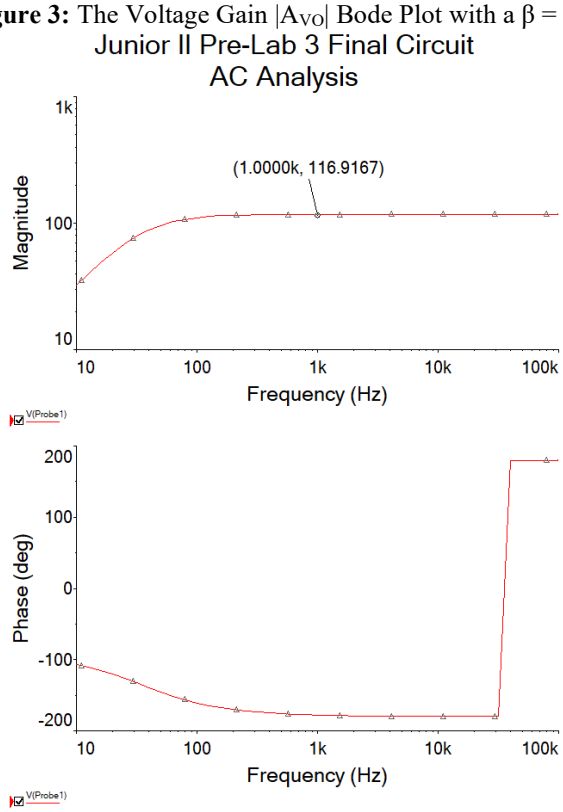

Figure 4: The Voltage Gain $|A_{VO}|$ Bode Plot with a $\beta = 250$

B. List of materials

- Active Components:
  - 2N2222A NPN Bipolar Junction Transistors
- Passive Components:
  - Resistors
  - Capacitors
- Power Supply & Signal Sources:
  - DC Power Supply
  - Function Generator
- Measurement Equipment:
  - Oscilloscope
  - Multimeter

C. Experimental Procedure

The experimental procedure involved a systematic approach to calculating and measuring key circuit parameters to verify the design and functionality of the constant current source.

First, the common-emitter stage was constructed using an equivalent resistance corresponding to the emitter follower's expected input impedance. The voltage gain, as well as key node voltages V_{B1}, V_{C1}, and V_{E1}, were measured, along with the collector current I_{C1} and emitter current I_{E1}. These values were obtained by first measuring the resistor values, then calculating the expected voltages and currents using circuit equations and finally verifying them through direct measurement with a multimeter.

After confirming the correct operation of the common-emitter stage, the emitter follower was built separately and tested using the same methodology. Once this circuit component was validated, the common-emitter and emitter-follower stages were combined to form the full current source. The complete circuit was then tested to measure the total voltage gain and source current. Additionally, the input resistance R_{in} and output resistance R_{out} of the full circuit were both calculated and measured to ensure the circuit met design requirements. These values were compared against theoretical predictions to evaluate the circuit's accuracy and performance. The results of the calculations and the simulation can be found in Table 1.

D. Results

The results of the calculations and the simulation can be found in Table 1.

Table 1: Calculated and Simulated Results

Variable	Simulated	Calculated	Measured
R_1	24kΩ	24kΩ	23.8652 kΩ
R_2	10 kΩ	10 kΩ	10.8755 kΩ
R_C	1.1kΩ	1.1 kΩ	1.11063 kΩ
R_{E1}	0.62 Ω	0.62 Ω	0.61575 kΩ
R_3	2 kΩ	2 kΩ	1.98283 kΩ
R_4	2.4 kΩ	2.4 kΩ	2.38619 kΩ
R_{E2}	1 kΩ	1 kΩ	0.97759 kΩ
R_L	10 kΩ	10 kΩ	9.96596 kΩ
V_{B1}	4.51 V	4.509208 V	4.494913 V
V_{C1}	7.93 V	6.825254 V	8.11178 V
V_{E1}	3.81 V	3.809208 V	3.83959 V

I_{C1}	6.11 mA	6.1454 mA	6.2021 mA
I_{E1}	6.76 mA	6.1864 mA	6.2357 mA
CE $\lvert A_V \rvert$	116	133.063124	114.882886
V_{B2}	8.15V	8.143155 V	8.159787 V
V_{E2}	7.5V	7.443155 V	7.518217 V
I_{C2}	7.40mA	7.5633 mA	7.6396 mA
I_{E2}	7.42mA	7.6137 mA	7.6905 mA
EF $\lvert A_V \rvert$	≅1	0.99651902	0.99852071
R_{in}	977 Ω	687.6157 Ω	814.9328 Ω
R_{out}	6.5 Ω	6.474848 Ω	6.733961 Ω
I_{source}	17.377mA	17.885 mA	16.780 mA
Total $\lvert A_V \rvert$	116	132.599933	114.244246

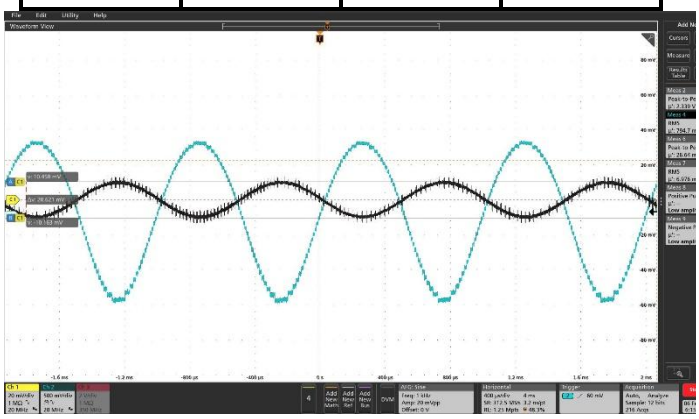

Figure 5: The Common Emitter Voltage Input/Output Waveform

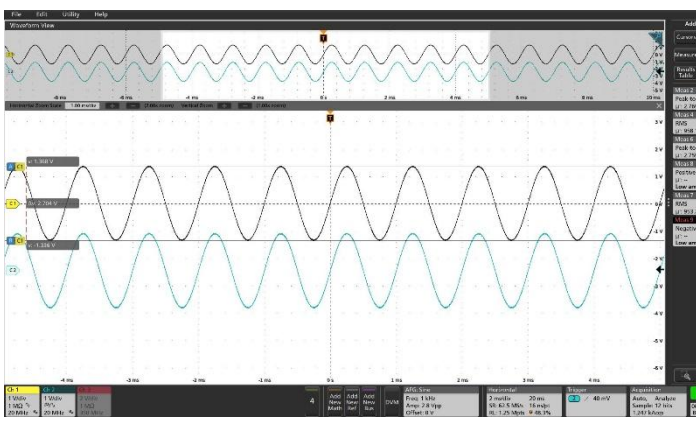

Figure 6: The Emitter Follower Voltage Input/Output Waveform

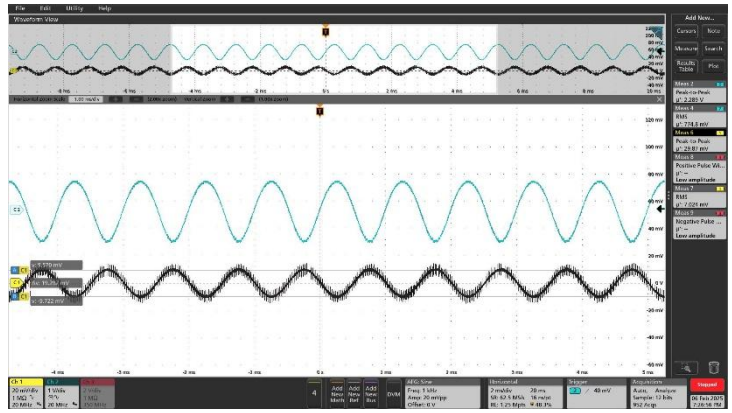

Figure 7: The Multi-stage Amplifier Voltage Input/Output Waveform

Figures 5, 6, and 7 show the waveforms created by the input and output voltage waveforms from peak to peak of the common emitter, emitter follower, and the full multi-stage amplifier circuit respectively. From these waveforms the difference in Voltage between the output and input can be calculated and analyzed.

E. Discussion of Results

The experiment successfully demonstrated the functionality of the designed circuit, with all specified requirements met. The measured voltage gain was found to be above 100, closely aligning with theoretical predictions. Additionally, the circuit's input and output resistances were within the expected range, confirming proper operation of both the common-emitter and emitter-follower stages.

The primary design requirements were:

1. A voltage gain exceeding 100.
2. An input resistance R_{in} greater than 600 Ω.
3. An output resistance R_{out} less than 8 Ω.
4. A source current less than 20 mA.

While the experiment met all design specifications, a slight discrepancy was observed between the calculated and measured voltage gain. The calculated gain was slightly higher than the measured value, which may be attributed to user error when determining voltage levels on the oscilloscope. Potential sources of error include probe placement, oscilloscope resolution limitations, and manual reading inaccuracies.

Despite these minor variations, the experiment confirmed the circuit's expected behavior, validating the theoretical design and demonstrating its effectiveness in maintaining stable performance within the given constraints.

F. Limitations

One limitation observed during the lab could be variations in transistor parameters, such as current gain (β), which directly affect the biasing conditions and small-signal performance. Temperature variations may also impact V_{BE}, leading to drift in operating points. Additionally, parasitic capacitances in the circuit and PCB wiring could introduce unwanted frequency-dependent behavior, affecting high-frequency response. Non-ideal resistor tolerances (typically ±5%) could cause slight deviations from the designed bias voltages. Furthermore,

loading effects from measurement instruments like the oscilloscope and multimeter might slightly alter circuit behavior, particularly in impedance-sensitive nodes. Lastly, power supply noise and thermal effects could introduce minor fluctuations in signal stability and distortion at higher gains.

IV. Conclusion

The experimental results confirm that the designed two-stage amplifier using 2N2222A BJTs successfully met the intended design specifications. The circuit demonstrated a high voltage gain, effective impedance matching, and proper frequency response, aligning well with theoretical predictions and simulation outcomes. The measured values closely matched the calculated parameters, validating the chosen biasing network and component selection. The use of a common-emitter amplifier for voltage gain and an emitter follower for impedance matching proved to be an efficient configuration, ensuring signal integrity and stable operation. Overall, this experiment provided valuable insights into multi-stage amplifier design, emphasizing the importance of bias stability, gain considerations, and impedance interactions in practical circuit implementations.

References

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